

4 The kidney is involved in the process of homeostasis in a mammal.

(a) Antidiuretic hormone (ADH) is required for homeostasis.

How many of these parts of the kidney respond to ADH?

- Bowman's capsule
- collecting duct
- loop of Henle
- proximal tubule

(1)

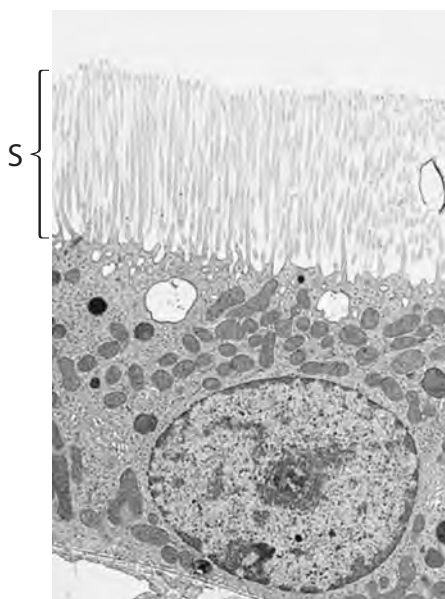
☐ **A** 1

☐ **B** 2

☐ **C** 3

☐ **D** 4

(b) The electron micrograph shows a transverse section of part of a proximal tubule.



(Source: <https://veteriankey.com/solute-reabsorption/>)

(i) Name the process by which the fluid moves from the plasma into the renal tubule (nephron).

(1)

(ii) Name the structures labelled S.

(1)

(c) A student compared the left renal veins of two individuals.

(i) The diameter of the left renal vein of one individual was 4.71 mm.

Calculate the cross-sectional area of this vein.

(2)

Answer mm²

(ii) The cross-sectional area of the left renal vein of the second individual was 0.194 cm².

Calculate the difference in the cross-sectional area in the left renal vein of these two individuals.

(1)

Answer

- (d) The table shows the concentration of glucose and urea in the blood plasma, glomerular filtrate and urine of an individual.

Substance	Concentration in blood plasma / g dm ⁻³	Concentration in glomerular filtrate / g dm ⁻³	Concentration in urine leaving collecting duct / g dm ⁻³
glucose	1.50	1.50	0.00
urea	0.30	0.30	21.00

Explain the concentration of glucose and urea in the urine.

Use the information in the table to support your answer.

(3)

[illegible]

(e) An individual produced 0.83 cm^3 of urine each minute.

Calculate the volume of urine produced by this individual in 24 hours.

Assume the urine production is at a constant rate.

Give your answer in dm^3 , to two significant figures.

(2)

Answer dm^3

(Total for Question 4 = 11 marks)